

Dhruva Bhat

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EDUCATION

University of California, Berkeley

B.S. in Electrical Engineering and Computer Science (EECS)

Berkeley, CA

Expected Graduation: May 2028

PROFESSIONAL EXPERIENCE

Oak Ridge National Lab | *Software Engineering Intern*

May 2026 - July 2026

- Designed and developed asynchronous FastAPI pipeline for drug-discovery applied molecular generation
- Generated over 700 molecular hypotheses per run using the Google Co-Scientist tournament-reasoning package
- Implemented retries and fallbacks for monitoring and observability capabilities during pipeline failures, reducing 6 manual handoffs to 1 request and developing 46 tests for orchestration and package failures, increasing reliability
- Refactored a C++ 14 speculative-simulation framework for cell-interaction modeling, making it usable for CPU-only and GPU-only builds and implemented 52 sanitizer tests, ensuring clean compilation and portability

Blueprint at Berkeley | *Contract Software Developer*

Feb 2026 - Present

- Developed a website for nonprofit Amigos De Los Rios, allowing for automatic volunteer route generation based on tree distances, volunteer availability, and days since watering, coordinating volunteers between over 300 sites
- Implemented backend and authentication for Vercel-deployed Next.js/Supabase platform, with authorization, row-level security, PostgreSQL migrations, and data access for locations, watering sessions, volunteers, and routes
- Developed a microservice for shortest routing algorithm for resource-based route generation, deployed on AWS Lambda, prioritizing 300+ API drawn locations and reducing staff labor from 20 to 2 hours per week

Carnegie Mellon University | *Software Engineer and AI Safety Researcher*

March 2026 - Present

- Collaborating with Professor Sarah H. Cen to study bounded rationality in LLMs, designing experiments that evaluate model decision-making against game-theoretic baselines across simulated institutional settings
- Developed a modular backend pipeline orchestrating large scale LLM in-game data gathering (17 GPU Hours Per Run) integrating OpenAI/HuggingFace/Ollama inference APIs, asynchronous experiment scheduling, and automated JSON Pipelines for data analysis, temperature-based performance differences with 96% confidence

University of California, San Francisco | *Machine Learning/Bioinformatics Researcher*

Aug 2025 - Present

- Working with Professor Oldham to identify optimal lung cancer gene markers, analyzing gene marker quantity in comparison to cell type prevalence across tissue samples, applying ML for big data analysis (8.68 million records)
- Analyzed genomic data to determine cancer markers, developing an unsupervised machine learning pipeline in R
- Ran pipeline on 125,000+ patient data rows, analyzing tissue data from lung cancer datasets by leveraging data cleaning, feature selection, hierarchical clustering, network-based outlier detection, and batch-effect correction
- Built a biological data scraper, pipeline, and neural network used for cross-study gene-marker identification

Garcia Center for Polymers at Engineering Interfaces | *Researcher*

Jun 2024 - Dec 2024

- Thin-Film Research: Enhanced prediction of semiconductor wafer thickness, correlating polystyrene molecular weight and film-thickness through 4D manifold learning in PyTorch and a Gaussian Process Regression model
- Hydrogel Research: Conducted rheological analysis of hydrogel (F88DMA cross-linked with APS, TEMED); analyzed data to determine 97% correlation between concentration and crosslinking time under various conditions

PROJECTS, PATENTS, PUBLICATIONS

Projects, Patents, and Publications | *EEG Vector Database, Opioid-Overdose Prevention Device, ML Polydispersity*

- Machine Learning for Assessing Impact of Polydispersity on Thin Film Thickness (MRS Communications)** - Developed PyTorch manifold-learning and Gaussian Process models for thin-film thickness prediction; published findings in MRS Communications and presented at MRS Boston 2024, APS Summit 2025
- EEG Vector Database** - Built a CEBRA-based representation-learning pipeline that converted simulated EEG signals into vector embeddings for comparing and distinguishing Parkinson's and healthy neural activity patterns
- Opioid-Overdose Prevention Device (Provisional Patent- U.S. 63/657,693)** - Developed an automated naloxone-delivery device using physiological monitoring, with a companion full-stack application and database

SKILLS

Technical: Python, C++, Javascript, Ansible, CI/CD, SQL, R, PyTorch, AWS Lambda, Supabase, PostgreSQL, OOP, Rest APIs, Docker, Vercel, Pandas, Scikit-learn, Git, Linux, LangGraph, RAG, Async I/O, ChromaDB, React

Relevant Coursework: Data Structures and Algorithms; Computer Architecture (Machine Structures); Efficient Algorithms and Intractable Problems; Discrete Mathematics and Probability; Signals, Dynamics, and Information